COMPUTER NETWORKS

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Lab Experiment – 1

1. Demo session of any two networking hardware and Functionalities.
2. Implement Byte Stuffing and Bit stuffing.
3. Execute networking commands with various options and update the result screens.

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NETWORKING DEVICES

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NETWORKING DEVICE

A networking device, also known as a network device, is a hardware component that connects computers, printers, servers, and other electronic devices to form a network. These devices enable communication and resource sharing between different networked devices. They play a crucial role in both small-scale home networks and large-scale enterprise networks. The networking devices discussed in this presentation are mentioned below:

1. **Modem:**

- Function: Converts digital data from a device into analog signals for transmission over communication lines and vice versa, enabling internet connectivity.
- Key Features: Modulation and demodulation, interfacing with ISP (Internet Service Provider), often includes a built-in router, supports various connection types (DSL, cable, fiber, satellite).

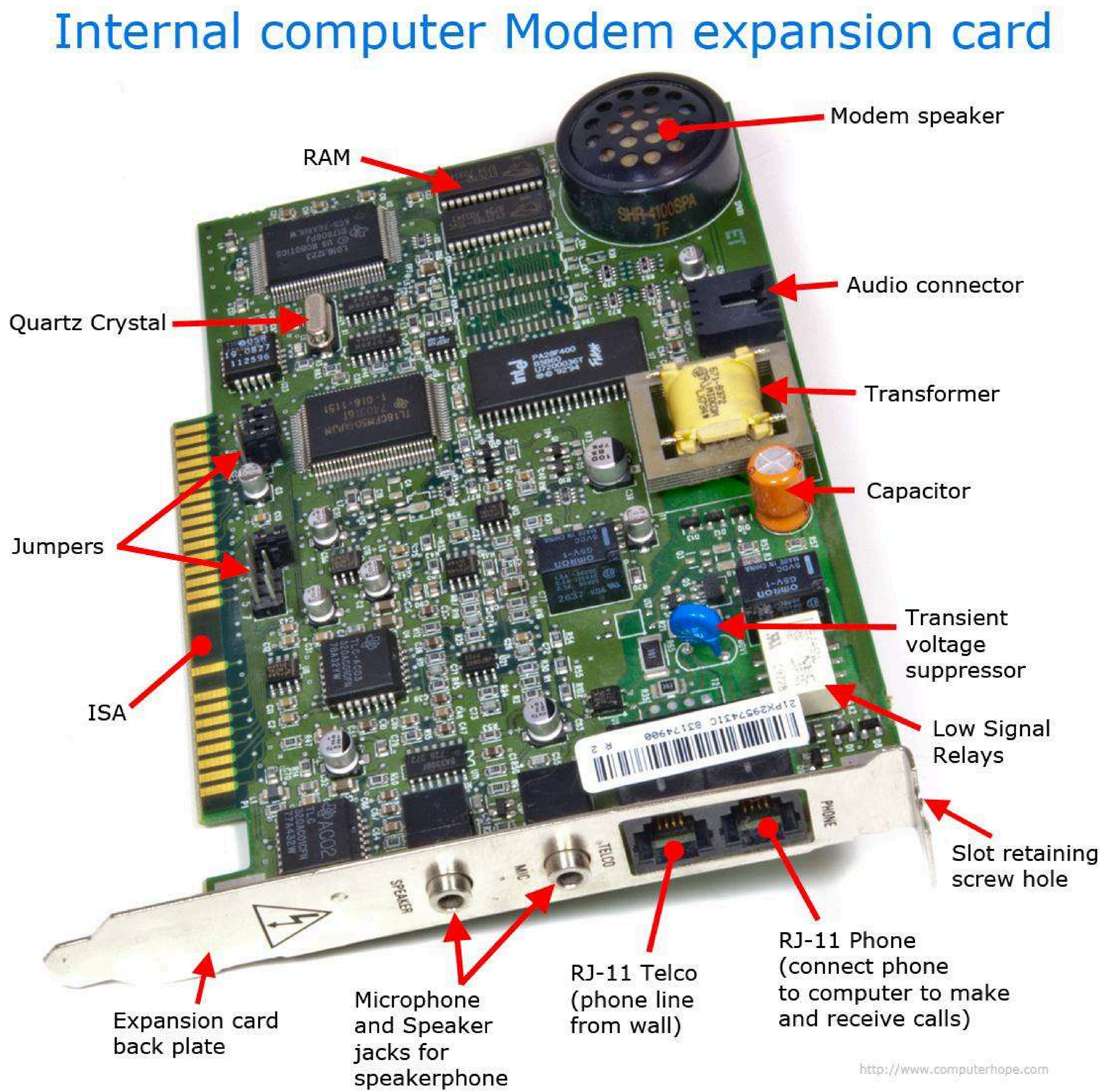
2. **Network Switch**

- Function: Connects multiple devices within the same network, allowing them to communicate directly by forwarding data to the correct destination device.
- Key Features: MAC address table, VLAN support, port mirroring.

MODEM & IT'S COMPONENTS

A SHORT INTRODUCTION

A modem (short for modulator-demodulator) is a hardware device that enables a computer or other device to connect to the internet. It converts digital data from a computer into analog signals that can be transmitted over telephone lines, cable systems, or satellite links, and vice versa



COMPONENTS OF MODEM

MODEM SPEAKER

- FUNCTION: ALLOWS USERS TO HEAR THE DIALING AND CONNECTION PROCESS.
- DETAILS: USEFUL FOR TROUBLESHOOTING CONNECTION ISSUES BY PROVIDING AUDIBLE FEEDBACK ON THE CONNECTION STATUS.

AUDIO CONNECTOR

- FUNCTION: CONNECTS THE MODEM TO AUDIO DEVICES.
- DETAILS: USED FOR VOICE COMMUNICATION OVER THE INTERNET, CONNECTING TO SPEAKERS OR MICROPHONES.

TRANFORMER

- FUNCTION: CONVERTS ELECTRICAL ENERGY TO THE APPROPRIATE VOLTAGE LEVELS.
- DETAILS: ENSURES THE MODEM’S COMPONENTS RECEIVE THE CORRECT VOLTAGE, PROTECTING THEM FROM POWER SURGES.

CAPACITOR

- FUNCTION: STORES AND RELEASES ELECTRICAL ENERGY.
- DETAILS: SMOOTHS OUT POWER FLUCTUATIONS AND FILTERS SIGNALS TO MAINTAIN STABLE MODEM OPERATION.

COMPONENTS OF MODEM

TRANSIENT VOLTAGE SUPPRESSOR (TVS)

- FUNCTION: PROTECTS THE MODEM FROM VOLTAGE SPIKES.
- DETAILS: ABSORBS AND DISSIPATES EXCESSIVE VOLTAGE TO PREVENT DAMAGE TO THE MODEM'S INTERNAL COMPONENTS, ENSURING STABLE OPERATION.

LOW SIGNAL RELAYS

- FUNCTION: MANAGE THE FLOW OF DATA SIGNALS.
- DETAILS: ELECTRICALLY CONTROLLED SWITCHES THAT DIRECT DATA SIGNALS WITHIN THE MODEM.

SLOT RETAINING HOLE

- FUNCTION: SECURES THE MODEM CARD WITHIN THE COMPUTER'S EXPANSION SLOT.
- DETAILS: ENSURES THE MODEM REMAINS FIRMLY IN PLACE, MAINTAINING A RELIABLE CONNECTION.

RJ-11 PHONE JACK

- FUNCTION: CONNECTS THE MODEM TO A TELEPHONE LINE.
- DETAILS: STANDARD CONNECTOR USED FOR ANALOG PHONE LINES, ALLOWING THE MODEM TO TRANSMIT AND RECEIVE DATA OVER A TELEPHONE NETWORK.

COMPONENTS OF MODEM

RJ-11 TELCO JACK

- FUNCTION: CONNECTS THE MODEM TO THE TELEPHONE COMPANY'S NETWORK.
- DETAILS: PROVIDES THE INTERFACE FOR THE MODEM TO COMMUNICATE WITH THE INTERNET SERVICE PROVIDER (ISP), ENABLING INTERNET CONNECTIVITY THROUGH THE PHONE LINE.

EXPANSION CARD BACK PLATE

- FUNCTION: PROVIDES A PHYSICAL INTERFACE FOR CONNECTING EXTERNAL CABLES.
- DETAILS: KEEPS THE MODEM SECURELY INSTALLED IN THE COMPUTER AND ALLOWS CONNECTION TO THE INTERNET.

QUARTZ CRYSTAL

- FUNCTION: PROVIDES A PRECISE CLOCK SIGNAL.
- DETAILS: ENSURES ACCURATE TIMING FOR DATA TRANSMISSION AND RECEPTION, CRUCIAL FOR MAINTAINING SYNCHRONIZATION.

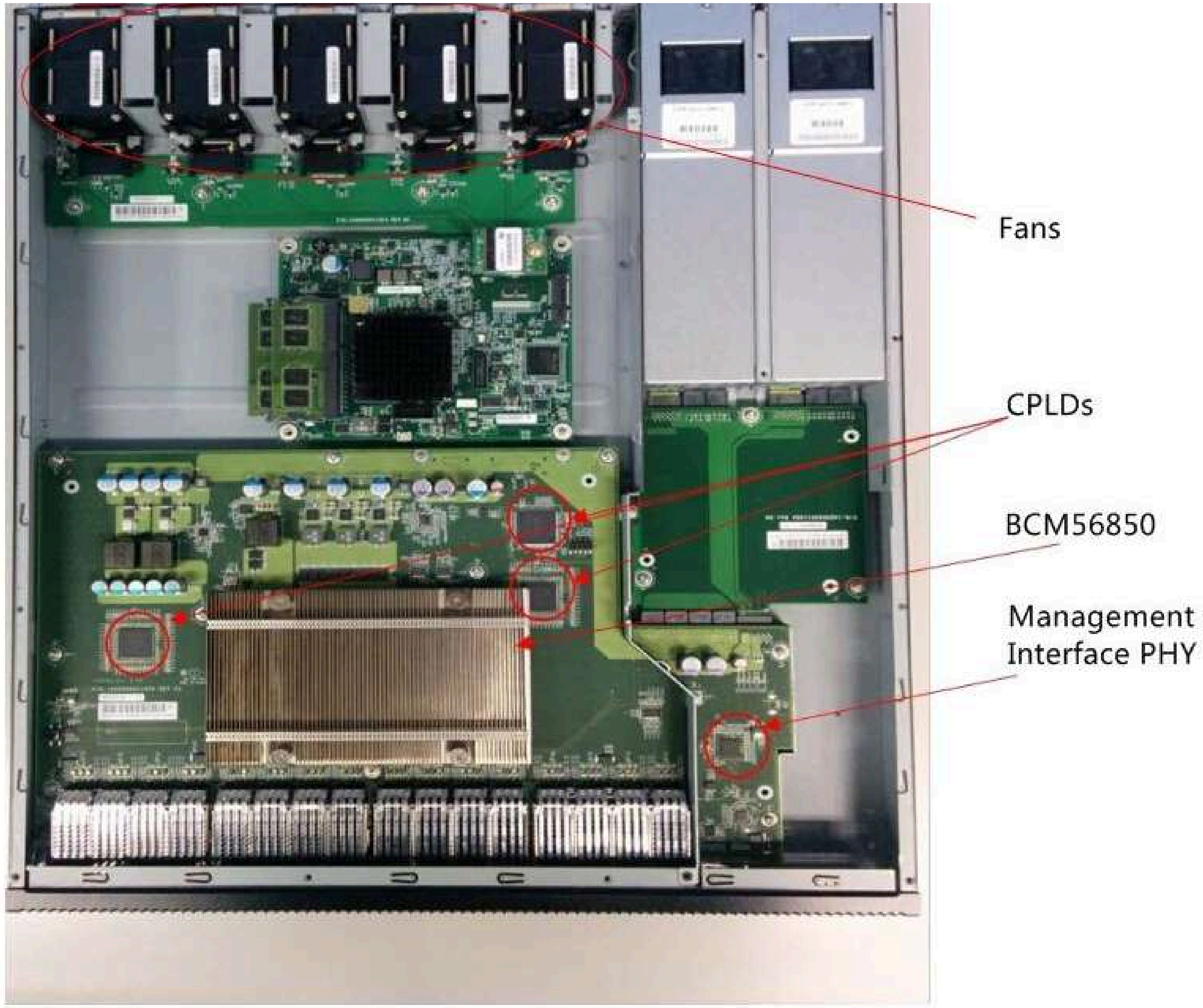
JUMPER

- FUNCTION: CONFIGURES SETTINGS ON THE MODEM.
- DETAILS: CREATES OR BREAKS ELECTRICAL CONNECTIONS ON THE CIRCUIT BOARD TO ADJUST MODEM SETTINGS.

SWITCH & IT'S COMPONENTS

A SHORT INTRODUCTION

A network switch is equipment that allows two or more IT devices, such as computers, to communicate with one another. Connecting multiple IT devices together creates a communications network. Compute, print, server, file storage, Internet access, and other IT resources can be shared across the network.



① 2 Power Supplies
② 5 Hot Swappble Fans
③ Intel C2538 CPU

COMPONENTS OF SWITCH

FANS

- FUNCTION: PROVIDE COOLING TO PREVENT OVERHEATING OF INTERNAL COMPONENTS.
- DETAILS: FANS CAN BE EITHER STANDARD OR HOT-SWAPPABLE. HOT-SWAPPABLE FANS ALLOW FOR REPLACEMENT OR MAINTENANCE WITHOUT SHUTTING DOWN THE DEVICE, ENSURING CONTINUOUS OPERATION.

CPLDS (COMPLEX PROGRAMMABLE LOGIC DEVICES)

- FUNCTION:IMPLEMENT CUSTOM LOGIC FUNCTIONS AND CONTROL TASKS WITHIN THE DEVICE.
- DETAILS: CPLDS ARE PROGRAMMABLE DEVICES USED TO PERFORM SPECIFIC HARDWARE FUNCTIONS. THEY CAN HANDLE TASKS LIKE SIGNAL ROUTING, PROTOCOL CONVERSION, AND CUSTOM DATA PROCESSING. THEY ARE MORE FLEXIBLE THAN FIXED-FUNCTION HARDWARE.

BCM56850 (BROADCOM SWITCH CHIP)

- FUNCTION:MANAGE AND ROUTE NETWORK TRAFFIC AT HIGH SPEEDS.
- DETAILS: THIS SWITCH CHIP IS DESIGNED FOR HIGH-PERFORMANCE NETWORK SWITCHING. IT SUPPORTS VARIOUS ETHERNET STANDARDS (E.G., 1G, 10G, 25G) AND ADVANCED FEATURES LIKE VLAN TAGGING, QUALITY OF SERVICE (QOS), AND NETWORK SECURITY. IT HELPS IN MANAGING MULTIPLE NETWORK INTERFACES AND ENSURES EFFICIENT DATA HANDLING.

COMPONENTS OF SWITCH

MANAGEMENT INTERFACE PHY

- FUNCTION:FACILITATE NETWORK MANAGEMENT AND MONITORING.
- DETAILS: THE PHY (PHYSICAL LAYER TRANSCEIVER) FOR THE MANAGEMENT INTERFACE PROVIDES CONNECTIVITY FOR OUT-OF-BAND MANAGEMENT. IT ALLOWS ADMINISTRATORS TO MONITOR, CONFIGURE, AND TROUBLESHOOT THE DEVICE THROUGH A DEDICATED NETWORK INTERFACE.

POWER SUPPLIES

- FUNCTION: SUPPLY ELECTRICAL POWER TO THE NETWORKING DEVICE.
- DETAILS:POWER SUPPLIES CONVERT AC POWER FROM THE WALL OUTLET TO THE DC POWER NEEDED BY THE DEVICE'S COMPONENTS. REDUNDANT POWER SUPPLIES ENSURE THAT THE DEVICE REMAINS OPERATIONAL IF ONE POWER SUPPLY FAILS. THEY ARE CRUCIAL FOR MAINTAINING DEVICE RELIABILITY AND UPTIME.

HOT SWAPPABLE FANS

- FUNCTION:ALLOW FOR THE REPLACEMENT OF FANS WITHOUT INTERRUPTING DEVICE OPERATION.
- DETAILS: HOT-SWAPPABLE FANS CAN BE REPLACED OR SERVICED WHILE THE DEVICE IS RUNNING, REDUCING MAINTENANCE DOWNTIME. THIS FEATURE IS IMPORTANT FOR HIGH-AVAILABILITY ENVIRONMENTS WHERE CONTINUOUS OPERATION IS CRITICAL.

COMPONENTS OF SWITCH

INTEL C2538 CPU

- FUNCTION:PERFORM MANAGEMENT AND CONTROL FUNCTIONS WITHIN THE NETWORKING DEVICE.
- DETAILS: THE INTEL C2538 IS AN ARM-BASED PROCESSOR KNOWN FOR ITS LOW POWER CONSUMPTION AND EFFICIENT PERFORMANCE. IT HANDLES TASKS SUCH AS DEVICE CONFIGURATION, MONITORING, AND RUNNING MANAGEMENT SOFTWARE. IT IS DESIGNED TO PROVIDE ADEQUATE PROCESSING POWER FOR NETWORK MANAGEMENT APPLICATIONS.

BROADCOM 56850 SWITCH CHIP

- FUNCTION: MANAGES NETWORK TRAFFIC AND SUPPORTS ADVANCED SWITCHING FEATURES.
- DETAILS:HIS CHIP IS CRUCIAL FOR HIGH-PERFORMANCE DATA HANDLING IN NETWORKING DEVICES. IT SUPPORTS MULTIPLE ETHERNET SPEEDS AND PROVIDES FEATURES LIKE HIGH-SPEED SWITCHING, TRAFFIC MANAGEMENT, AND SUPPORT FOR COMPLEX NETWORK PROTOCOLS.

→

2. BYTE STUFFING

Code:

```
#include <iostream>
#include<string>
using namespace std;
int main(){
    string inp; int len=0;
    getline(cin,inp);
    cout<< "after stuffing"<<endl;
    char afterstuffing[100];
    afterstuffing[0]='$';
    len=1;
    for (int i=0;i<inp.length();i++){
        if(inp[i]=='@'|| inp[i]=='$'){
            afterstuffing[len]=inp[i];
            len++;
            afterstuffing[len]='@';
            len++;
        }
        else{
            afterstuffing[len++]=inp[i];
        }
    }
    afterstuffing[len]='$';
    len++;

    for (int i=0;i<len;i++){
        cout<<afterstuffing[i];
    }
    cout<<endl<<"after unstuffing"<<endl;
    char unstuffing[100];
    int len2=0;
    for (int i=len-1;i>0;i--){
        if(afterstuffing[i]=='@'){
            i--;
        }
        unstuffing[len2]=afterstuffing[i];
        len2++;
    }
    for (int i=len2-1;i>0;i--){
```

```
            cout<<unstuffing[i];
        }
        cout<<endl;
    }
```

Sample Output

```
(env) karnwalpoorva23@Apoorvas-MacBook-Air codeforces % cd "/Users/karnwalpoorva23/Desktop/code
 && "/Users/karnwalpoorva23/Desktop/codeforces/"tempCodeRunnerFile
$
after stuffing
$$@$
after unstuffing
$%
```

```
(env) karnwalpoorva23@Apoorvas-MacBook-Air codeforces % cd "/Users/karnwalpoorva23
karnwalpoorva23/Desktop/codeforces/"bytestuffing
ABC
after stuffing
$ABC$
after unstuffing
ABC%
```

```
karnwalpoorva23/Desktop/codeforces/"bytestuffing
ABC@$
after stuffing
$ABC@@$@$
after unstuffing
ABC@$
```

```
(env) karnwalpoorva23@Apoorvas-MacBook-Air codeforces % cd "/Users/karnwalpoo
karnwalpoorva23/Desktop/codeforces/"bytestuffing
$$@@$
after stuffing
$$@$@@@@$@$
after unstuffing
$$@@$
```

3. NETWORKING COMMANDS:

1. Ping Command

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Batch1> ping google.com

Pinging google.com [142.250.67.142] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 142.250.67.142:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
Control-C
PS C:\Users\Batch1> ping vtop.vit.ac.in

Pinging vtop.vit.ac.in [10.10.5.168] with 32 bytes of data:
Reply from 10.10.5.168: bytes=32 time=1ms TTL=61
Reply from 10.10.5.168: bytes=32 time=2ms TTL=61
Reply from 10.10.5.168: bytes=32 time=1ms TTL=61
Reply from 10.10.5.168: bytes=32 time=1ms TTL=61

Ping statistics for 10.10.5.168:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
PS C:\Users\Batch1> ping vit.ac.in

Pinging vit.ac.in [10.10.7.35] with 32 bytes of data:
Reply from 10.10.7.35: bytes=32 time=1ms TTL=252
Reply from 10.10.7.35: bytes=32 time=2ms TTL=252
Reply from 10.10.7.35: bytes=32 time=2ms TTL=252
Reply from 10.10.7.35: bytes=32 time=2ms TTL=252

Ping statistics for 10.10.7.35:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

2. Tracert Command

```
PS C:\Users\Batch1> tracert google.com

Tracing route to google.com [142.250.67.142]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  10.30.157.1
  2     1 ms     1 ms     1 ms  10.30.0.5
  3     1 ms     1 ms     1 ms  10.11.0.2
  4     1 ms     1 ms     1 ms  10.10.100.201
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *
PS C:\Users\Batch1> tracert vit.ac.in

Tracing route to vit.ac.in [10.10.7.35]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  10.30.157.1
  2    26 ms     1 ms     1 ms  10.30.0.5
  3     1 ms     2 ms     2 ms  10.11.0.2
  4     3 ms     2 ms     2 ms  10.10.7.201
  5     2 ms     2 ms     2 ms  vit.ac.in [10.10.7.35]

Trace complete.
```

3. netstat Comand

```
PS C:\Users\Batch1> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.30.157.17:13116     bom07s31-in-f10:https  CLOSE_WAIT
  TCP    10.30.157.17:38043     123:https              CLOSE_WAIT
  TCP    10.30.157.17:44674     bom07s29-in-f3:https   CLOSE_WAIT
  TCP    10.30.157.17:46362     ns31661457:6568        SYN_SENT
  TCP    127.0.0.1:49686        SJT417SCOPE007:49688   ESTABLISHED
  TCP    127.0.0.1:49687        SJT417SCOPE007:49689   ESTABLISHED
  TCP    127.0.0.1:49688        SJT417SCOPE007:49686   ESTABLISHED
  TCP    127.0.0.1:49689        SJT417SCOPE007:49687   ESTABLISHED
  TCP    127.0.0.1:49690        SJT417SCOPE007:49691   ESTABLISHED
  TCP    127.0.0.1:49691        SJT417SCOPE007:49690   ESTABLISHED
```

4. getmac Command

```
PS C:\Users\Batch1> getmac

Physical Address    Transport Name
=================== ==========================================================
00-50-56-C0-00-01   \Device\Tcpip_{F8CB1245-9D6E-4BCE-811E-052C00095A56}
00-50-56-C0-00-08   \Device\Tcpip_{5D270B27-E7D3-4592-9B48-BD44C76EEFE5}
7C-57-58-CB-EC-51   \Device\Tcpip_{C93FB492-CC1D-41F5-A2A5-0DA392280B0E}
```

5. hostname Command

```
PS C:\Users\Batch1> hostname
SJT417SCOPE007
```

6. ipconfig Command

```
PS C:\Users\Batch1> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3439:a476:8a1b:8127%16
   IPv4 Address. . . . . . . . . . . : 10.30.157.17
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.30.157.1

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b6c0:a136:e303:be15%17
   IPv4 Address. . . . . . . . . . . : 192.168.40.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::445:be0c:e30f:f080%9
   IPv4 Address. . . . . . . . . . . : 192.168.242.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

7. nslookup

```
PS C:\Users\Batch1> nslookup
Default Server:  umbrella2.vit.ac.in
Address:  10.10.4.10
```

8. arp -a Comand

```
PS C:\Users\Batch1> arp -a 10.30.157.1

Interface: 10.30.157.17 --- 0x10
  Internet Address      Physical Address      Type
  10.30.157.1           f8-e5-7e-bb-52-61     dynamic
```

```
PS C:\Users\Batch1> arp -a

Interface: 192.168.242.1 --- 0x9
  Internet Address      Physical Address      Type
  192.168.242.254       00-50-56-e0-aa-ec     dynamic
  192.168.242.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 10.30.157.17 --- 0x10
  Internet Address      Physical Address      Type
  10.30.157.1           f8-e5-7e-bb-52-61     dynamic
  10.30.157.11          7c-57-58-cb-df-a9     dynamic
  10.30.157.14          7c-57-58-cb-ec-81     dynamic
  10.30.157.16          7c-57-58-cb-df-4b     dynamic
  10.30.157.19          7c-57-58-cb-df-e4     dynamic
  10.30.157.21          7c-57-58-cb-df-54     dynamic
  10.30.157.23          7c-57-58-cb-ec-4e     dynamic
  10.30.157.25          7c-57-58-cb-ec-31     dynamic
  10.30.157.26          7c-57-58-cb-ec-3c     dynamic
  10.30.157.27          7c-57-58-cb-df-9b     dynamic
  10.30.157.29          7c-57-58-cb-ec-64     dynamic
  10.30.157.30          7c-57-58-cb-df-10     dynamic
  10.30.157.32          7c-57-58-cb-ec-0b     dynamic
  10.30.157.33          7c-57-58-cb-df-c4     dynamic
  10.30.157.34          7c-57-58-cb-ec-69     dynamic
  10.30.157.36          7c-57-58-cb-df-6f     dynamic
  10.30.157.37          7c-57-58-cb-df-d3     dynamic
  10.30.157.38          7c-57-58-cb-ec-39     dynamic
  10.30.157.39          7c-57-58-cb-eb-33     dynamic
  10.30.157.40          7c-57-58-cb-df-70     dynamic
  10.30.157.42          7c-57-58-cb-df-58     dynamic
  10.30.157.43          7c-57-58-cb-eb-7d     dynamic
  10.30.157.44          7c-57-58-cb-df-33     dynamic
  10.30.157.46          7c-57-58-cb-ec-3e     dynamic
  10.30.157.48          7c-57-58-cb-df-84     dynamic
  10.30.157.49          7c-57-58-cb-df-69     dynamic
```

9. **nbtstat:** to know the details of NetBIOS over TCP/IP

```
PS C:\Users\Batch1> nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

10.systeminfo Command

```
PS C:\Users\Batch1> systeminfo

Host Name:                 SJT417SCOPE007
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.22631 N/A Build 22631
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Member Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          SJT417SCOPE00
Registered Organization:
Product ID:                00331-20090-00000-AA583
Original Install Date:     31-07-2023, 17:55:54
System Boot Time:          26-07-2024, 07:37:40
System Manufacturer:       HP
System Model:              HP Pro Tower 280 G9 PCI Desktop PC
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 151 Stepping 2 GenuineIntel ~2100 Mhz
BIOS Version:              AMI F.24, 17-11-2023
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              00004009
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     16,069 MB
Available Physical Memory: 10,414 MB
Virtual Memory: Max Size:  18,501 MB
Virtual Memory: Available: 11,937 MB
Virtual Memory: In Use:    6,564 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    VITUNIVERSITY.LOCAL
Logon Server:              \\ADC-SCOPE
Hotfix(s):                 4 Hotfix(s) Installed.
                           [01]: KB5039895
                           [02]: KB5027397
                           [03]: KB5040442
                           [04]: KB5039338
Network Card(s):           3 NIC(s) Installed.
                           [01]: VMware Virtual Ethernet Adapter for VMnet1
                                 Connection Name: VMware Network Adapter VMnet1
                                 DHCP Enabled:    Yes
```